

OAKLAND COUNTY INTL, MI, USA

WMO: 726375

Lat: 42.665N

Lon: 83.418W

Elev: 976

StdP: 14.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 0.3	(c) 4.8	(d) -7.9	(e) 3.7	(f) 1.7	(g) -2.5	(h) 5.0	(i) 6.7	(j) 26.2	(k) 24.9	(l) 24.0	(m) 24.3	(n) 7.0	(o) 290	(p) 0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	89.7	72.8	86.5	71.0	83.8	69.6	75.3	84.4	73.6	82.2	72.1	80.4	10.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.5	124.8	80.3	71.1	118.7	78.5	69.6	112.9	77.1	39.4	85.2	37.8	82.0	36.4	80.1	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	20.3	18.3	-6.1	93.0	5.9	2.5	-10.3	94.8	-13.8	96.3	-17.1	97.7	-21.4	99.5
			-6.7	78.5	5.8	1.6	-10.9	79.7	-14.2	80.6	-17.5	81.5	-21.7	82.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.8	23.6	25.7	34.5	47.1	58.4	68.3	72.4	70.7	63.6	51.4	39.4	29.1
	DBStd	19.34	11.29	10.77	10.83	8.96	8.61	6.72	5.70	5.37	7.42	8.39	8.94	9.52
	HDD50	3245	818	680	498	157	20	0	0	0	2	84	336	650
	HDD65	6600	1282	1100	947	540	244	42	8	14	113	428	769	1113
	CDD50	2806	1	1	16	70	279	548	693	640	410	129	17	2
	CDD65	685	0	0	1	3	38	140	236	189	71	7	0	0
	CDH74	5645	0	0	8	41	376	1205	2032	1378	551	54	0	0
	CDH80	1648	0	0	1	4	91	378	663	357	145	9	0	0
(14) Wind	WSAvg	8.6	9.7	9.6	9.7	10.0	8.7	7.7	7.2	6.6	7.0	8.3	9.1	9.3
(15) Precipitation	PrecAvg	33.7	2.2	1.9	2.2	2.8	3.8	3.4	3.3	3.1	3.1	3.0	2.6	2.2
	PrecMax	42.6	3.9	3.9	3.9	6.0	8.4	5.4	6.8	5.7	7.3	7.9	4.5	3.7
	PrecMin	23.7	0.7	0.7	0.7	0.3	1.6	1.4	1.2	0.5	0.7	0.4	0.8	0.8
	PrecStd	4.8	0.9	0.9	0.9	1.2	1.5	1.2	1.5	1.2	1.7	1.5	1.1	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.6	58.7	72.9	79.5	86.8	91.5	93.3	91.3	89.0	80.5	67.6	57.5
		MCWB	52.4	51.4	60.4	62.5	69.4	73.8	74.4	73.7	71.2	68.2	57.1	54.1
	2%	DB	49.4	51.3	64.0	73.0	82.4	88.3	89.8	87.4	84.0	73.9	61.4	51.9
		MCWB	46.9	46.8	54.4	58.5	66.4	71.7	73.3	71.6	69.4	62.9	53.8	48.7
	5%	DB	43.4	45.2	57.1	68.1	78.8	84.4	86.5	83.8	80.6	70.0	57.2	46.0
		MCWB	40.1	40.2	49.6	55.1	64.8	69.2	71.4	70.2	67.1	61.2	50.9	42.1
	10%	DB	38.0	39.4	51.7	63.2	73.9	81.3	83.6	81.3	76.4	65.5	53.6	41.5
		MCWB	35.6	35.6	45.2	52.6	61.9	67.8	70.0	68.8	65.5	57.9	48.7	38.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	54.1	61.5	65.2	71.8	76.6	78.4	77.3	74.5	70.1	59.5	54.9
		MCDB	54.6	56.7	71.6	75.9	82.8	88.2	88.7	87.9	84.0	78.0	63.4	57.7
	2%	WB	46.9	46.8	56.3	61.1	69.1	73.7	75.8	74.5	71.8	66.1	55.8	49.3
		MCDB	48.2	50.4	62.4	69.3	78.6	84.1	84.8	82.2	79.5	71.5	60.6	51.5
	5%	WB	40.4	40.5	51.2	57.4	66.8	71.9	74.1	72.7	69.7	62.1	52.2	42.8
		MCDB	43.1	44.5	56.6	65.8	75.3	81.3	82.5	80.1	76.1	67.9	56.2	45.7
	10%	WB	35.9	36.1	44.8	53.6	64.0	70.1	72.3	71.0	67.6	58.9	48.3	38.5
		MCDB	38.3	39.5	50.4	61.8	71.9	78.1	80.3	78.4	74.1	64.5	52.9	41.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	14.5	16.5	19.2	19.3	18.8	18.6	17.6	18.6	16.7	13.6	11.2
		MCDBR	17.0	20.3	24.5	26.5	23.7	22.0	21.1	19.9	22.6	22.6	19.7	15.9
	5% WB	MCWBR	15.2	16.4	16.8	15.8	12.1	9.8	9.0	8.5	10.1	13.1	14.6	13.7
		MCWBR	16.8	19.5	22.7	23.3	20.9	18.9	18.2	16.9	18.3	19.1	18.0	15.4
(40) Clear-Sky Solar Irradiance	taub		0.293	0.327	0.343	0.386	0.412	0.434	0.424	0.417	0.380	0.351	0.315	0.300
		taud	2.389	2.275	2.366	2.267	2.244	2.214	2.262	2.270	2.360	2.432	2.490	2.418
	Ebn at Noon		269	274	285	279	274	267	268	266	269	264	260	255
		Edh at Noon	25	33	34	41	43	44	42	40	34	28	23	23
(44) All-Sky Solar Radiation	RadAvg		480	756	1121	1459	1700	1919	1929	1662	1339	864	535	378
	RadStd		43	93	116	128	148	118	92	65	122	77	69	30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page